

Seeing the Invisible

Physics-Guided Missing-Physics Learning for PEMFC Flooding Prediction

Gang-Hyun Shin, Jong-Jin Jang, Geon-Ho Shin, Je-Hyeon Ryu, Seo-Yeon Kim, Min-Seo Baek, Min-Ji Lee, Chan-Seo Lee, Kee-Youn Yoo*
Department of Chemical and Biomolecular Engineering, Seoul National University of Science and Technology

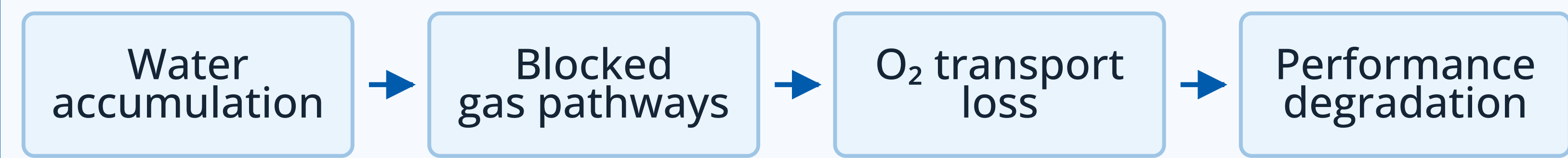


ABSTRACT

Liquid water accumulation in porous gas diffusion layers limits mass transport in polymer electrolyte membrane fuel cells, while uncertain constitutive relations remain a major source of prediction error. We propose a physics-informed grey-box framework that retains the governing equations in a finite-element solver and learns only the missing gas-phase transport closure. The closure is identified from synthetic reference responses using discrete-adjoint sensitivities without prescribing its functional form. After training at RH 50%, the learned closure is frozen and evaluated zero-shot at unseen RH 70% on 16×40 and 32×80 meshes. The recovered law reaches 2.16% relative L^2 error, while saturation-field errors remain below 0.56%. Field-level agreement is retained without retraining, supporting transfer across the tested humidity and discretization changes and establishing a foundation for physics-guided digital twins of PEMFC flooding.

INTRODUCTION

Why flooding matters



What limits flooding prediction?

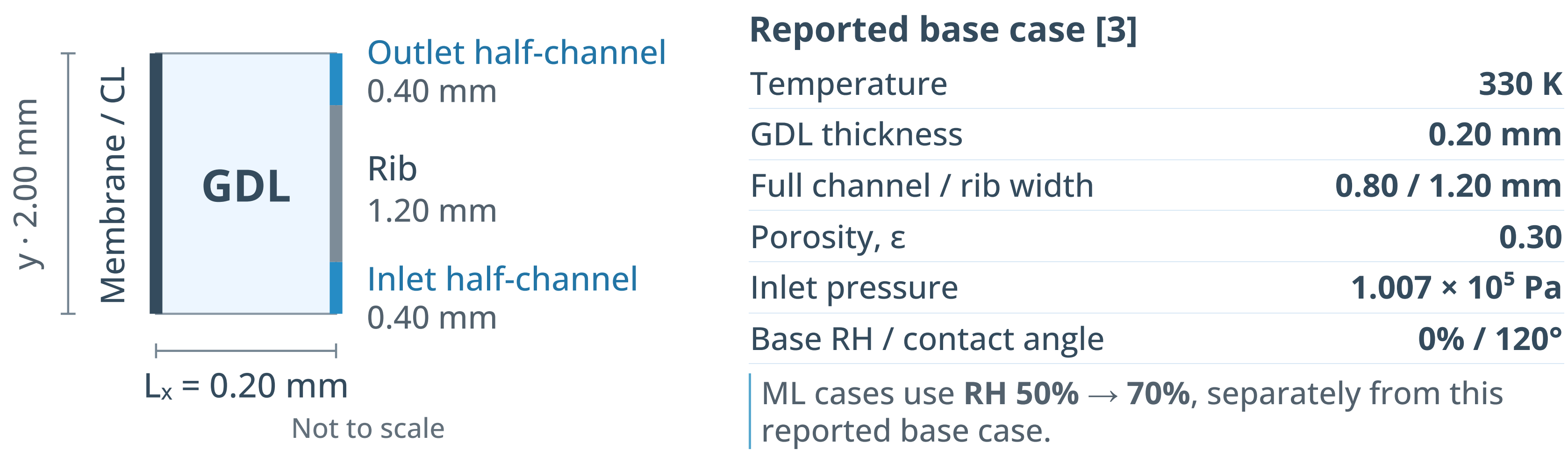
Observation	Internal liquid-water saturation $s(x,y)$ cannot be directly and continuously observed during operation.
Modeling	Flooding predictions depend on uncertain constitutive relations such as $D_{\text{eff}}(s)$, $k_r(s)$, and $p_c(s)$.

Existing ML vs our learning objective

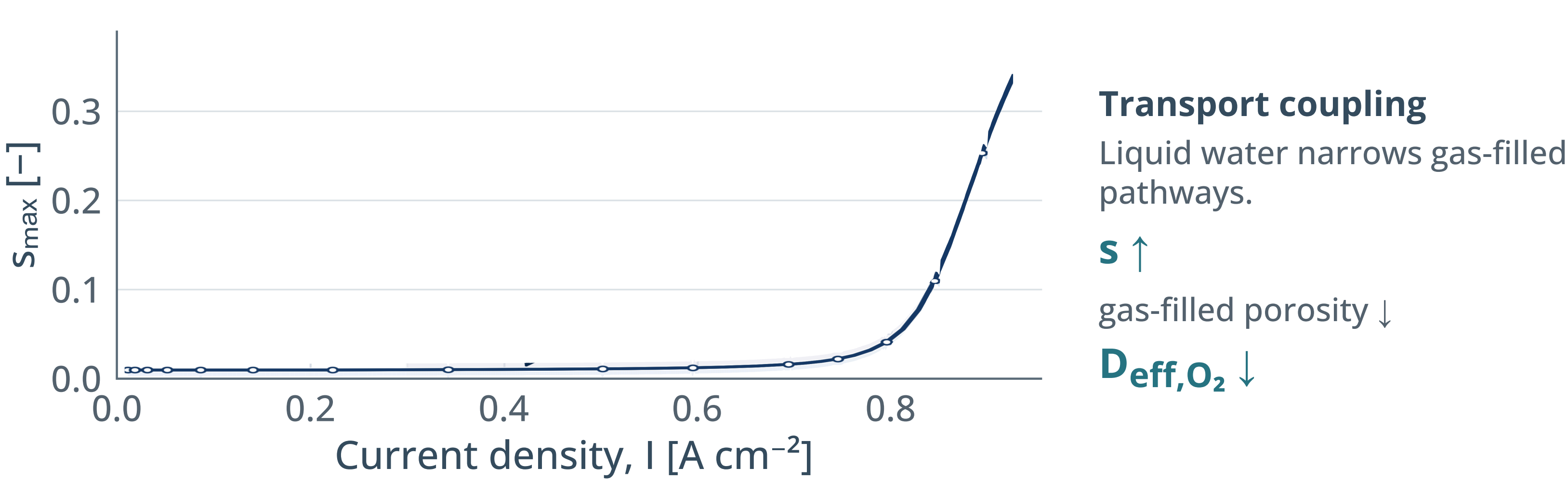
PINN Learn the PDE solution Physics is enforced through equation-residual training.	FNO / Neural operator Learn the solution operator Map input fields or parameters to solution fields.	OUR APPROACH FEM + Missing Physics ML Learn the constitutive law Only the uncertain transport relation is learned.
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RESULTS

01 Model domain & reference conditions



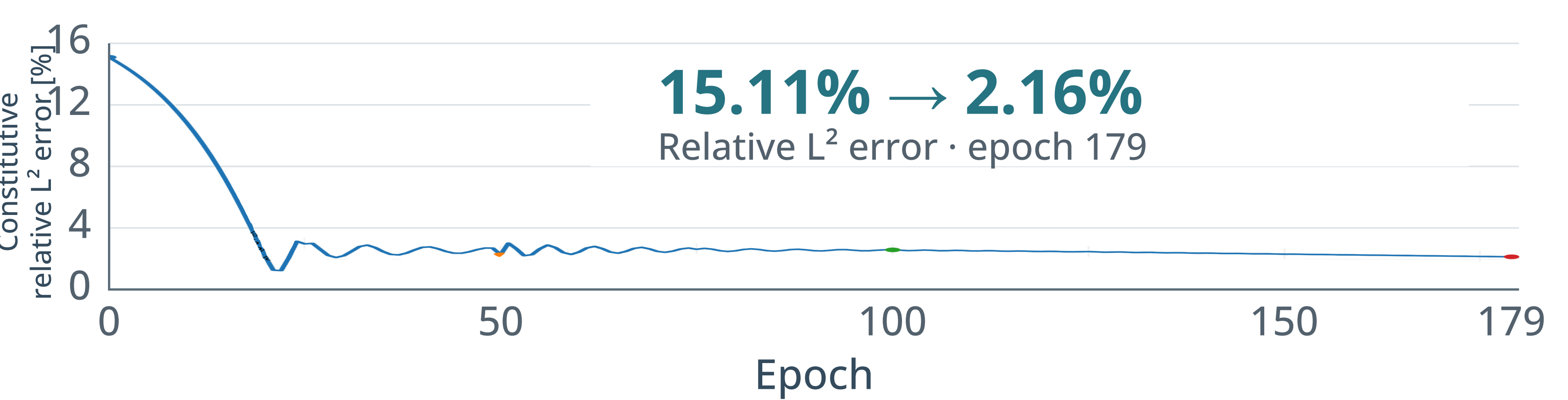
02 Water accumulation grows at high current



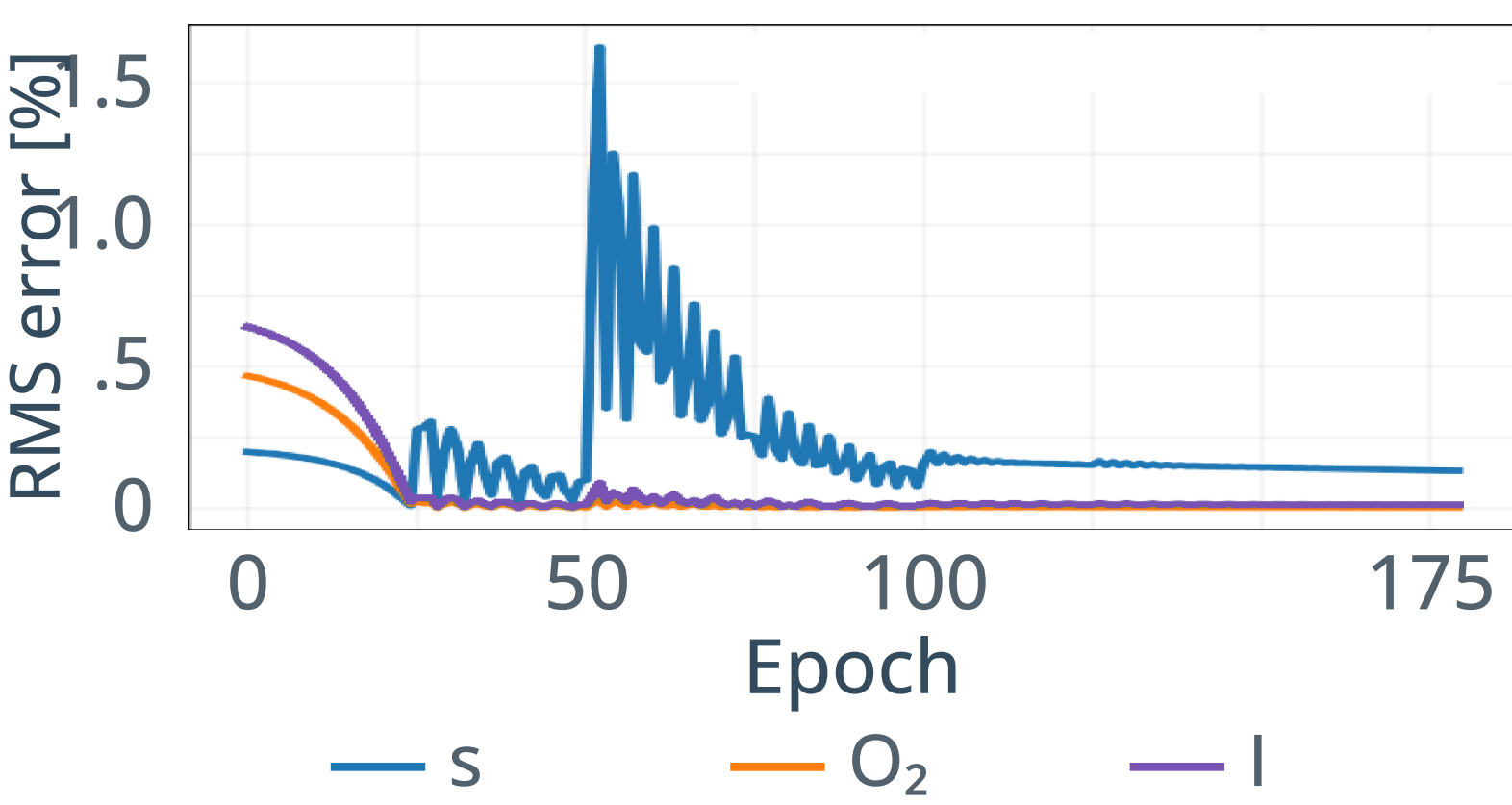
s_{max} : maximum liquid saturation within the GDL.

03 Recover the missing transport relation

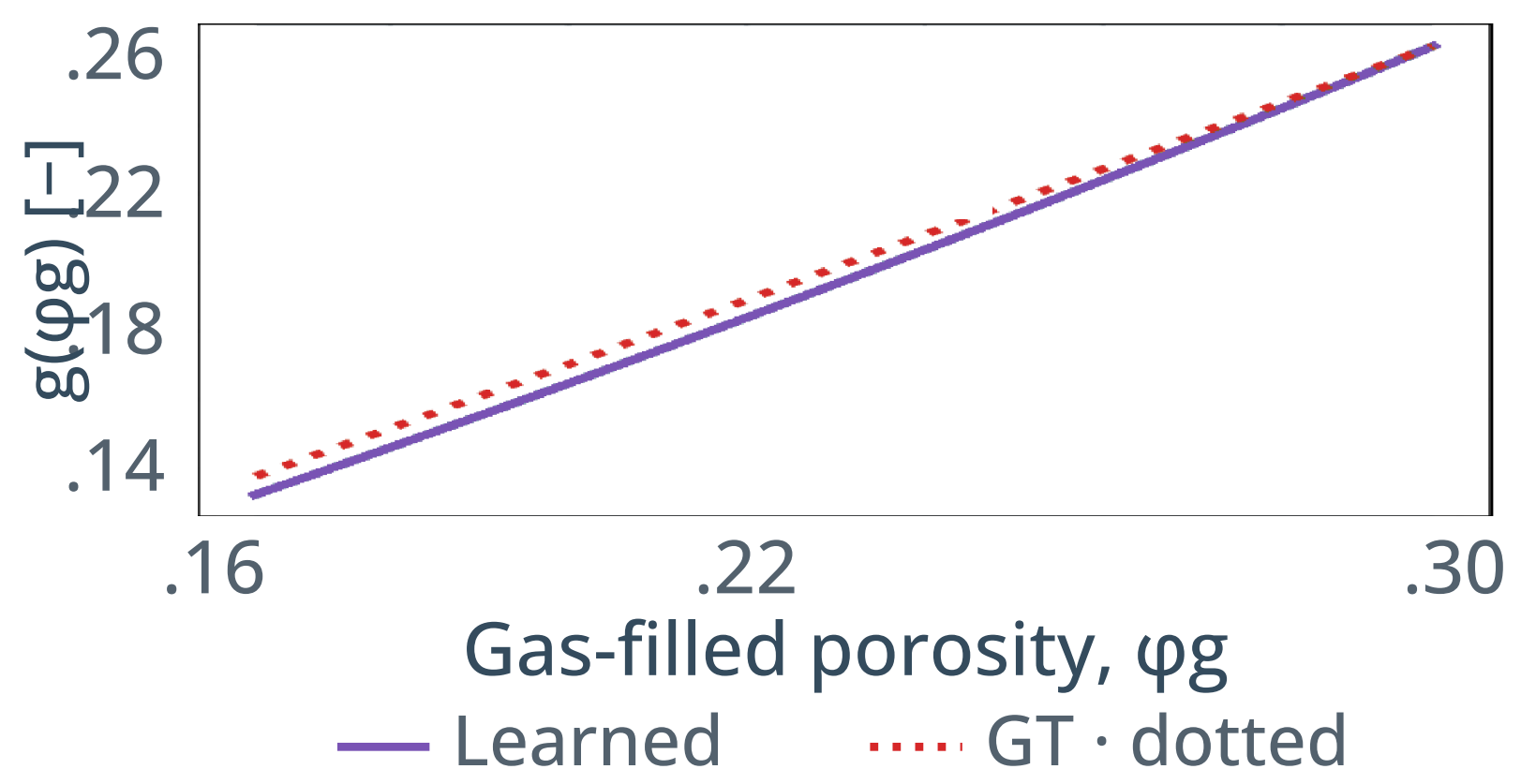
Train at RH 0%: infer the gas-phase closure from synthetic FEM responses.



State/response reconstruction

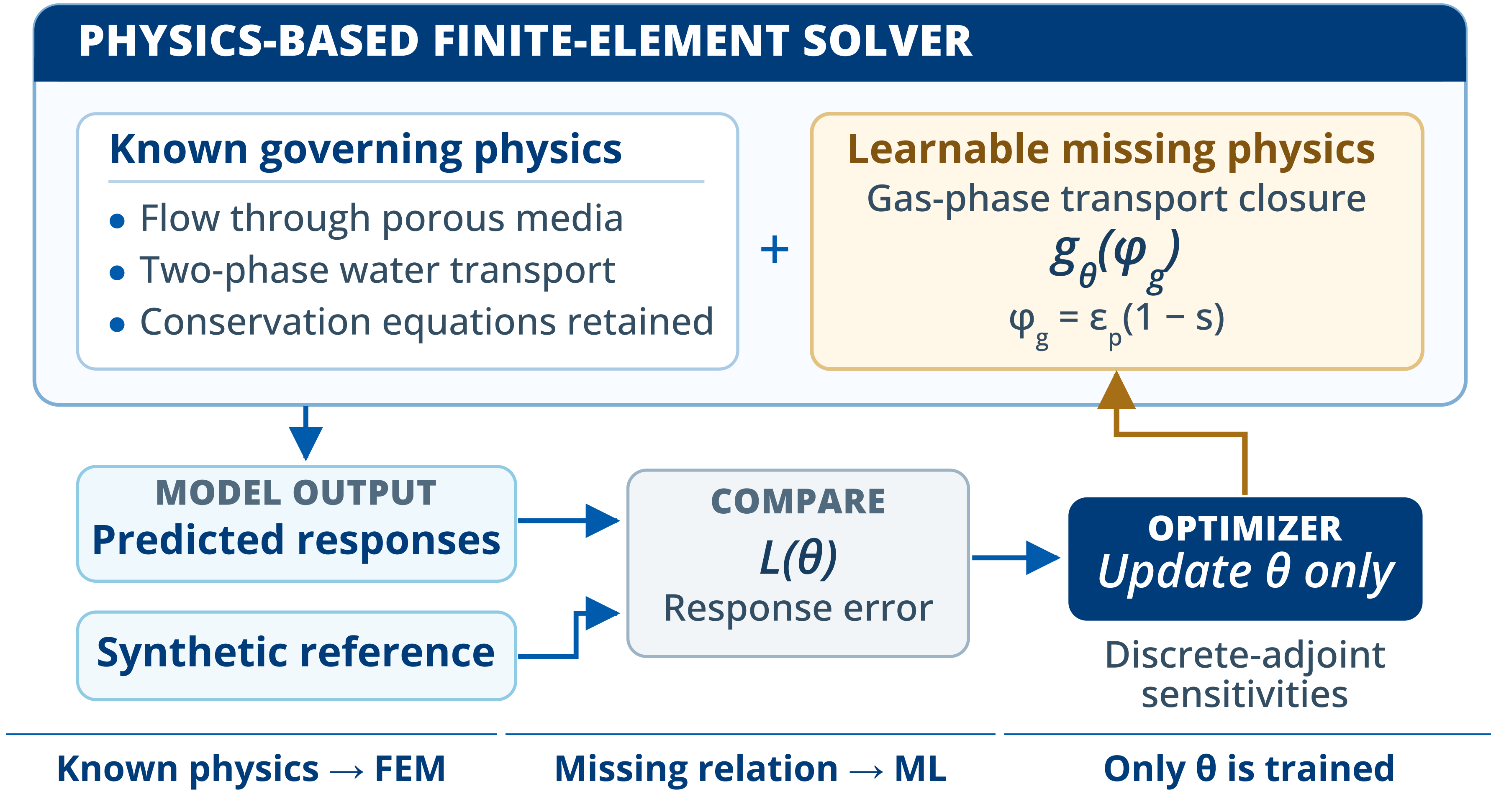


Recovered constitutive law

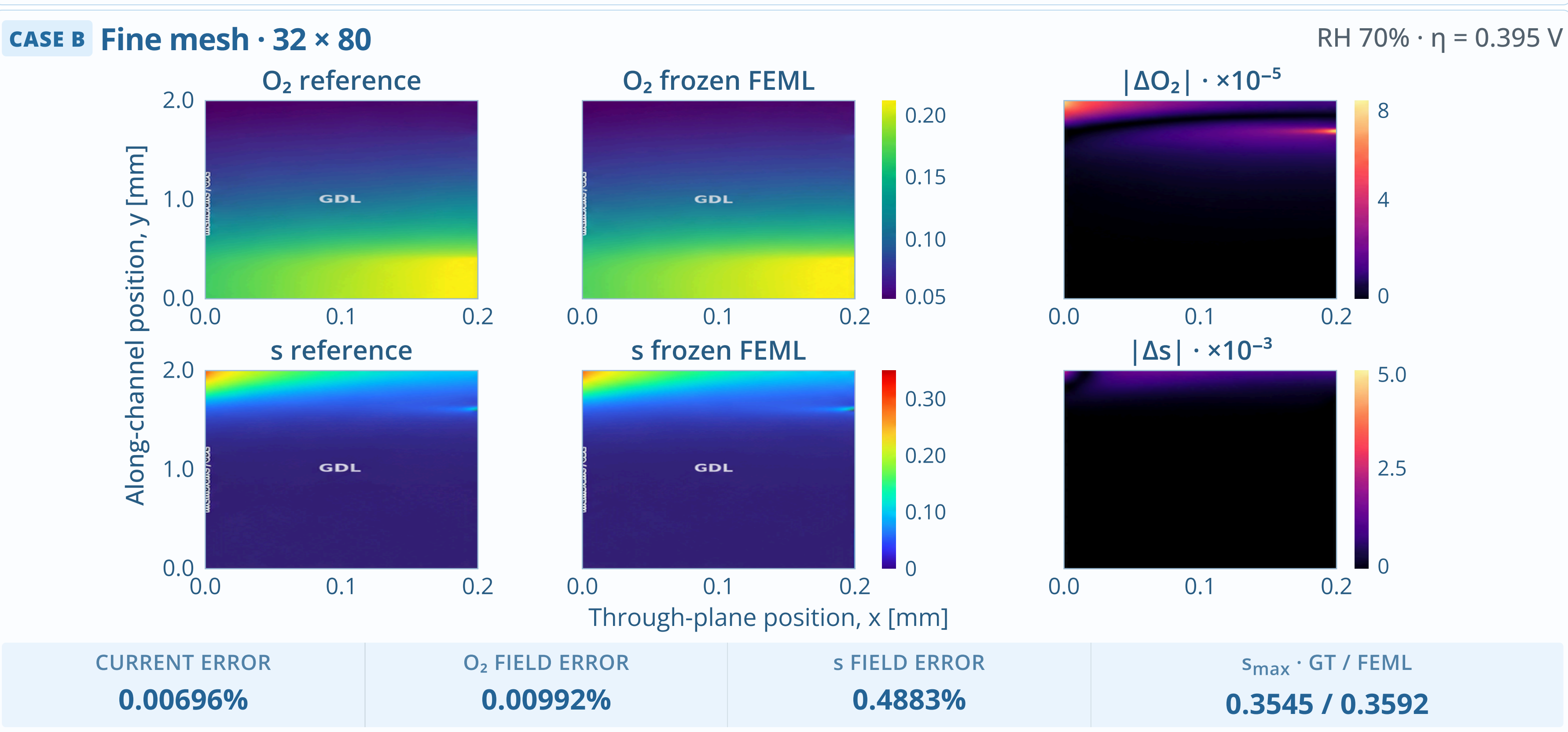
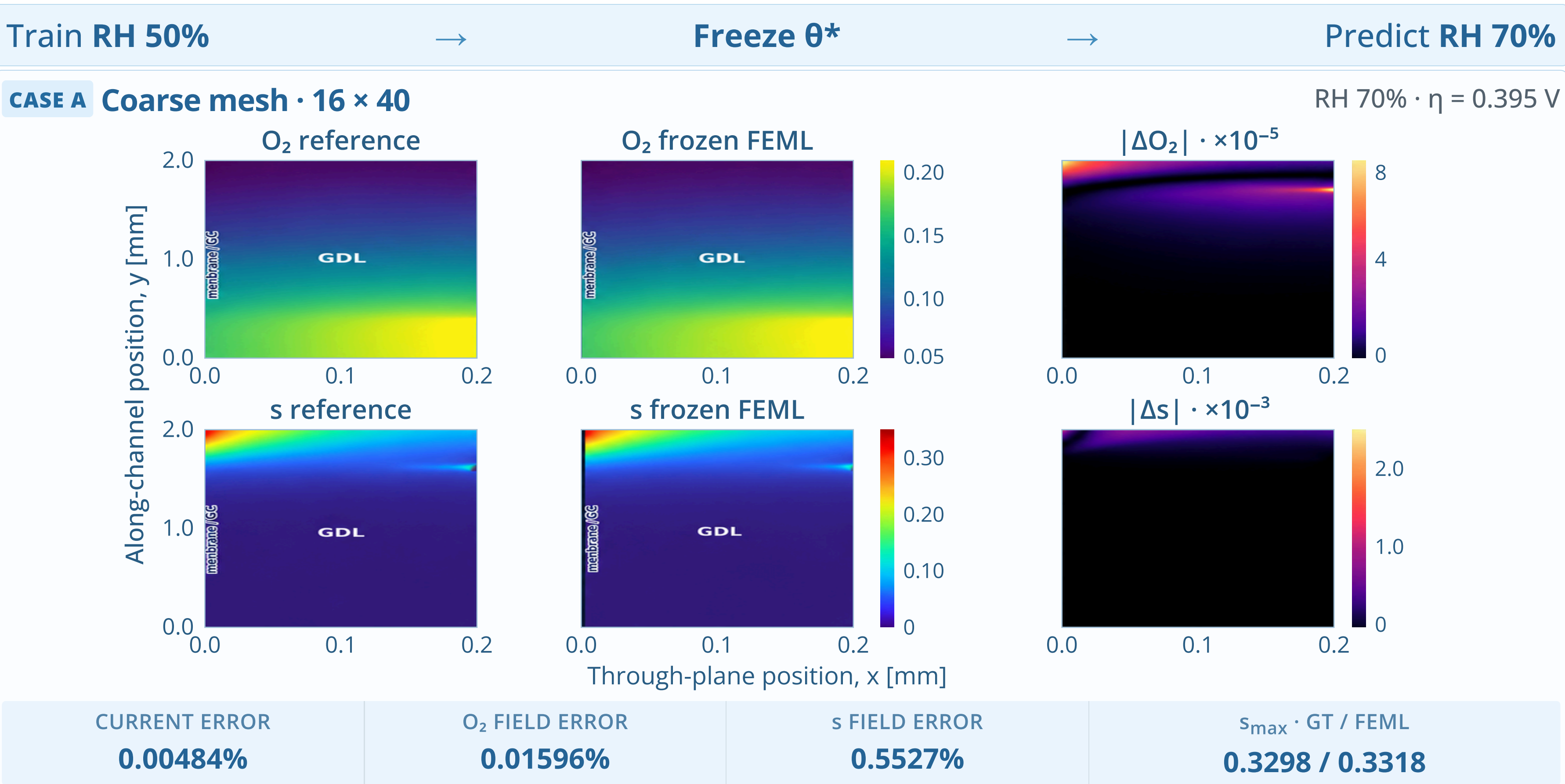


$$D_{\text{eff}} = D_{\text{dry}} g_{\theta}(\phi_g) \quad \phi_g = \epsilon_p(1 - s) \quad \text{GT: } g(\phi_g) = \phi_g^{\tau}$$

MODELING FRAMEWORK



04 Zero-shot transfer with a frozen closure



CONCLUSIONS & OUTLOOK

A learned transport law can be reused inside FEM without retraining.

01 Physically interpretable learning. The recovered closure links gas-filled porosity to transport, making the learned relation directly inspectable.

02 Transfer within the tested conditions. Agreement at changed RH and discretization supports reuse of the closure; it does not establish mesh convergence.

03 Next step: predictive validation. Test against experimental responses and a wider operating range, then assess closure identifiability and uncertainty.

REFERENCES [1] A. Farsi, N. Bouziani, D. A. Ham. Missing physics discovery through fully differentiable finite element-based machine learning. Scientific Reports (2026).
[2] M. Abdollahzadeh et al. Analysis of PEM fuel cell cathode two-dimensional modeling. Energy 68, 478–494 (2014).